# cyclepoint – brain

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## Load packages

```
library(dplyr)
library(ggplot2)
library(mgcv)
library(gamm4)
library(ggeffects)
```

## Load data

```
structural <- read.csv("structural.csv")
structural$region <- as.factor(structural$region)
```

### Structural MRI analysis

Includes data from an open-source dense-sampling dataset of a single individual (Pritschet et al., 2020).

#### Run model

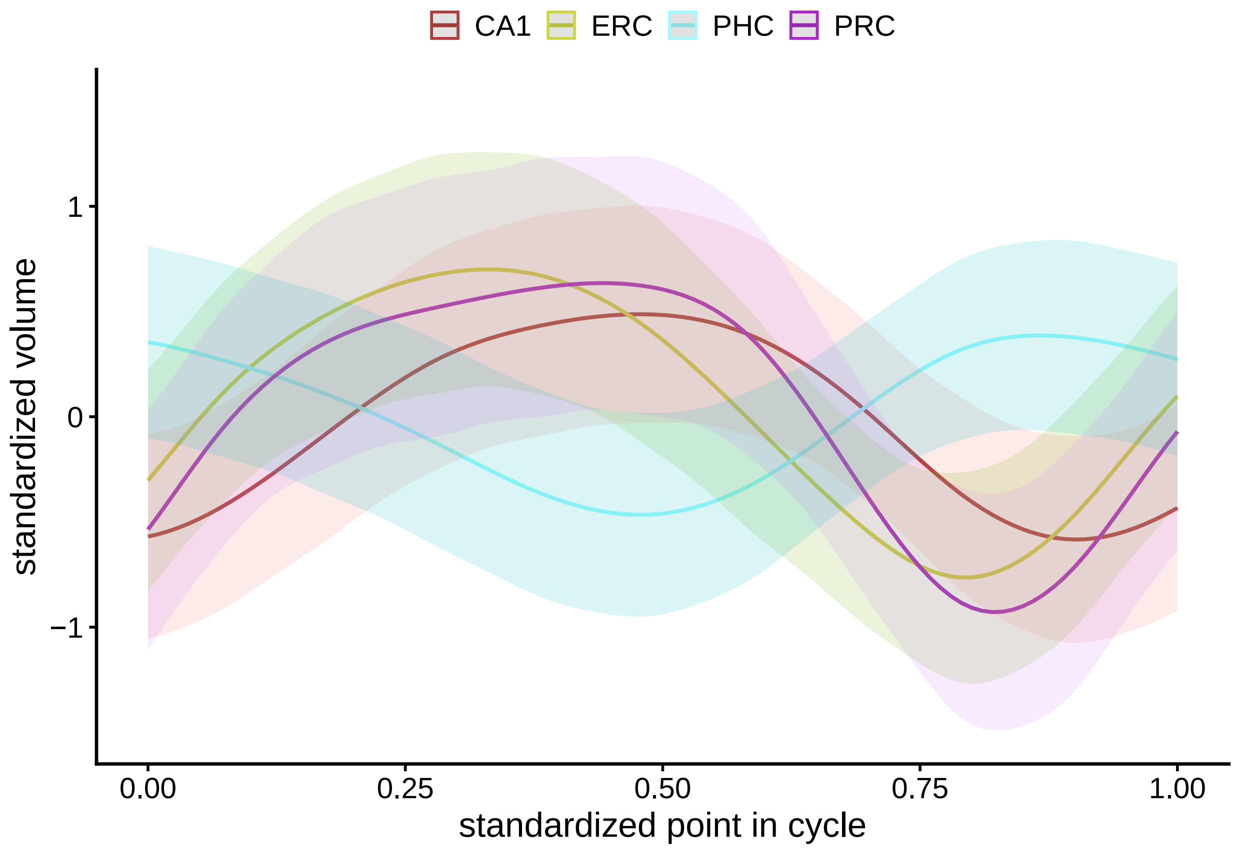
```
cyc <- paste('cyclepoint [',toString(seq(0.0,1.0,by=0.01)),']')

mghpc <- gamm4(value~s(cyclepoint,bs='cc',by=region) + ICVz,
               data=structural)
summary(mghpc$gam)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## value ~ s(cyclepoint, bs = "cc", by = region) + ICVz
##
## Parametric coefficients:
```

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.073e-11  6.335e-02   0.000    1.000
## ICVz        -6.508e-02  6.484e-02  -1.004    0.317
##
## Approximate significance of smooth terms:
##                                   edf Ref.df     F p-value
## s(cyclepoint):regionca1_s  1.976e+00      8 0.978 0.01922 *
## s(cyclepoint):regionca23_s 4.996e-07      8 0.000 0.43930
## s(cyclepoint):regiondg_s   1.025e+00      8 0.203 0.22527
## s(cyclepoint):regionerc_s  2.328e+00      8 1.717 0.00197 **
## s(cyclepoint):regionphc_s  1.732e+00      8 0.654 0.05338 .
## s(cyclepoint):regionprc_s  2.757e+00      8 1.864 0.00229 **
## s(cyclepoint):regionsub_s  1.044e-07      8 0.000 0.53236
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =   0.16
## lmer.REML = 557.53  Scale est. = 0.81462   n = 203
```

### Plot results

```r
plot(ggpredict(mghpc$gam,terms=c(cyc,'region [ca1_s, erc_s, phc_s, prc_s]'))) +
  labs(x='standardized point in cycle',y='standardized volume')+
  coord_cartesian(xlim=c(0,1),ylim=c(-1.5,1.5))+
  scale_color_manual(values = c("#ac3f3b", "#ced645", "#9efaff", "#a828c5"),
                     labels = c("CA1", "ERC", "PHC","PRC")) +
  theme_classic()+
  theme(axis.line=element_line(linewidth=0.6),
        axis.text=element_text(size=11),
        axis.title=element_text(size=13),
        plot.title = element_blank(),
        legend.position='top',
        legend.title = element_blank(),
        legend.text=element_text(size=11),
        legend.margin=margin(t=0,unit='cm'),
        legend.spacing.x=unit(1,'mm'),
        legend.key.size=unit(4,'mm'))
```



CA1
ERC
PHC
PRC
standardized volume
1
0
−1
0.00
0.25
0.50
0.75
1.00
standardized point in cycle